LiteDepthwiseNet: A Lightweight Network for Hyperspectral Image Classification

Benlei Cui, Xue-Mei Dong, Qiaoqiao Zhan, Jiangtao Peng, and Weiwei Sun, *Senior Member, IEEE*

***Abstract*— Deep learning methods have shown considerable potential for hyperspectral image (HSI) classification, which can achieve high accuracy compared with traditional methods. However, they often need a large number of training samples and have a lot of parameters and high computational overhead. To solve these problems, this article proposes new network architecture, LiteDepthwiseNet, for HSI classification. Based on 3-D depthwise convolution, LiteDepthwiseNet can decompose standard convolution into depthwise convolution and pointwise convolution, which can achieve high classification performance with minimal parameters. Moreover, we remove the ReLU layer and batch normalization layer in the original 3-D depthwise convolution, which is likely to improve the overfitting phenomenon of the model on small-sized data sets. In addition, focal loss is used as the loss function to improve the model's attention on difficult samples and unbalanced data, and its training performance is significantly better than that of cross-entropy loss or balanced cross-entropy loss. Experiment results on five benchmark hyperspectral data sets show that LiteDepthwiseNet achieves state-of-the-art performance with a very small number of parameters and low computational cost.**

***Index Terms*— Depthwise convolution, focal loss, hyperspectral image (HSI) classification, lightweight network.**

I. Introduction

IN RECENT years, as a kind of optical remote sensing image with high spectral resolution, hyperspectral images (HSIs) have attracted much attention because of their unique properties and the massive information they contain [1]. HSIs have been widely used in mineral exploration [2], soil salinity estimation [3], and anomaly detection [4]. At present, HSI classification is one of the most active research areas in the remote sensing field. Its goal is to classify each marked pixel in the image correctly, where a pixel can be regarded as a high-dimensional vector with each entry corresponding to the spectral reflectance in a specific wavelength [5].

Early work on HSI classification mainly focused on exploring spectral feature information and ignoring spatial information [6]–[10], which often leads to poor generalization performance. After this, some studies began to incorporate spatial contextual information into pixel-level classifiers. Representative work includes extended morphological profiles (EMPs) [11], [12], edge-preserving filtering [13], multiscale covariance maps [14], local covariance matrix representation [15] and sparse representation models [16]–[20], etc. Due to the spatial homogeneous characteristics of HSI, spectral-spatial HSI classification methods usually show better results than spectral-based methods. Notwithstanding, most of spectral-spatial methods depend on hand-crafted or shallow features, which are usually designed for specific tasks and rely on expert knowledge [21]. This largely limits the flexibility and applicability of these methods in difficult situations.

Recently, deep learning (DL) methods have been applied in HSI classification successfully. The earlier DL-based HSI classification methods based on fully connected neural networks, such as stacked autoencoders (SAEs) [22] and recursive autoencoders (RAEs) [23], can only handle 1-D vectors. Thus, the spatial structure information of an HSI is destroyed. The emergence of convolutional neural networks (CNNs) overcomes this defect. Several 2-D CNN-based architectures were proposed, including R-VCANet [24], Bayesian 2-D CNN [25] and squeeze multibias network (SMBN) [26], etc. However, an HSI has too many channels, which often causes 2-D convolution kernels to be too deep, and there is also a significant increase in the number of parameters. Therefore, 3-D CNN-based HSI classification methods were proposed. Lee and Kwon [27] described a contextual deep CNN by applying multiple 3-D local convolutional filters with different sizes to jointly exploit the spatial and spectral features of an HSI. Chen *et al.* [28] established a 3-D CNN-based feature extraction model with regularization to extract the effective spectral-spatial features of HSIs. Hamida *et al.* [29] proposed a new 3-D DL method that can simultaneously process spectral and spatial information while incurring lower computation cost (i.e., floating point operations, FLOPs). Although 3-D CNN can reduce the number of parameters, it still incurs more calculational consumption because it needs to traverse

Manuscript received September 22, 2020; revised December 17, 2020 and January 18, 2021; accepted February 21, 2021. Date of publication March 11, 2021; date of current version December 9, 2021. This work was supported in part by the First Class Discipline of Zhejiang—A (Zhejiang Gongshang University —Statistics); in part by the National Natural Science Foundation of China under Grant 11701509, Grant 61871177, Grant 11771130, and Grant 41971296; in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR19D010001; in part by the Open Fund of State Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, under Grant 18R05; and in part by the National Key Research and Development Program of China under Grant 2020YFA0714200. *(Corresponding author: Xue-Mei Dong.)*

Benlei Cui, Xue-Mei Dong, and Qiaoqiao Zhan are with the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou 310018, China (e-mail: dongxuemei@zjgsu.edu.cn).

Jiangtao Peng is with the Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Weiwei Sun is with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: sunweiwei@nbu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3062372

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

in depth and lacks a global view of spectral information. HybridSN, a recent work proposed by Roy *et al.* in [30], combined 2-D CNN and 3-D CNN, where 3-D CNN was used to mine spatial and spectral information and 2-D CNN integrated the information to output the prediction result. It can effectively improve classification accuracy and has a global view to integrate spectral features. However, without an end-to-end architecture, principal component analysis (PCA) was still used to compress the redundant spectral information, which adds computational cost. Besides, a cascaded recurrent neural network model using gated recurrent units for HSIs classification was proposed in [31]. It has smaller numbers of parameters to train than CNNs and can fully tap the redundant and complementary information of HSIs. By integrating features extracted from CNNs and graph convolutional networks (GCNs), a mini-batch GCN was developed in [32]. It achieves better classification results in HSIs with regards to the single CNN or GCN models.

Although the aforementioned DL models have shown excellent HSI classification performance, they usually need a large number of training samples and network parameters and also have higher computational cost. For HSI classification, the available labeled pixels are usually very limited due to the difficulty in collecting and cost in labeling. Moreover, the distribution of categories in these labeled data is imbalanced. Constructing robust DL models with high performance and low computational cost based on imbalanced and small sample-sized data is a huge challenge in this field. Recently, a lightweight 3-D CNN network, the asymmetric inception network (AINet), was proposed in [33]. It emphasized spectral signatures rather than spatial contexts and used a data fusion transfer learning strategy for a better model initialization and saving the training time. However, it does not perform well in small-sized samples. The double-branch dual-attention mechanism network (DBDA) given in [34] captured spectral and spatial features separately. And the channel-wise attention mechanism and spatial-wise attention mechanism were adopted to refine the feature maps, which result in good accuracy on several HSI data sets. Later, a lightweight neural network, LiteDenseNet [35], was proposed. Different from traditional 3-D CNN architectures, LiteDenseNet has a 3-D two-way dense structure and uses group convolution, which greatly reduces the number of parameters and computational cost. However, the group convolution cuts off the connection of different channels and may lead to a loss of accuracy. To solve this problem and further reduce the parameters, we propose a lightweight network, LiteDepthwiseNet, for HSI classification tasks. Its main advantages are summarized as follows.

1) A 3-D depthwise convolution is used to replace the group convolution. The pointwise convolution in the 3-D depthwise convolution can connect all hyperspectral channels, and the corresponding network architecture has a full-channel receptive field and is more suitable for the HSI classification task. Based on this new network architecture, LiteDepthwiseNet only involves a very small number of parameters and relatively small FLOPs.

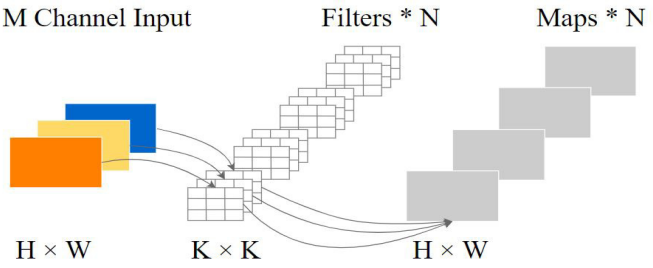


Fig. 1. Standard convolution.

2) Rather than using the mainstream cross-entropy loss (CEL), our proposed method uses focal loss (FL) as the loss function. The FL helps to improve the performance for most data sets under this new network framework since it sufficiently mines the difficult-to-classify samples in HSI images.
3) The middle activation layer and normalization layer in the original 3-D depthwise convolution network are stripped, which can reduce the complexity of the model, so our model is likely to avoid the overfitting phenomenon on HSI data sets with very few training samples.

The rest of this article is arranged as follows. In Section II, we briefly introduce the related work. The structure of the proposed LiteDepthwiseNet is detailed in Section III. The comparison experiment results of nine algorithms on five well-known HSI data sets are reported in Section IV. Section V summarizes this article.

II. Related Work

This section describes some related concepts and gives a brief introduction to LiteDenseNet, which inspires the main design of the proposed network architecture in this article.

A. 2-D Depthwise Separable Convolution

Different from standard convolution, the depthwise separable convolution proposed in [36] includes a depthwise convolution, i.e., a spatial convolution performed independently over each channel of an input, followed by a pointwise convolution, i.e., a 1×1 convolution, projecting the channels output by the depthwise convolution onto a new channel space. Figs. 1 and 2 show a simple schematic of a standard convolution and a depthwise separable convolution, respectively.

By splitting the standard convolution into two processes of depthwise convolution and pointwise convolution, the computational cost and the number of parameters in the 2-D convolution can be significantly reduced under certain conditions.

B. Cross Entropy Loss

The CEL function is a loss function often used in classification problems. It is an important concept in information theory, where it is mainly used to measure the difference between two probability distributions.

The standard CEL for classification problems is,

$$\mathrm{CEL}(p, y) = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1-p), & \text{otherwise} \end{cases} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

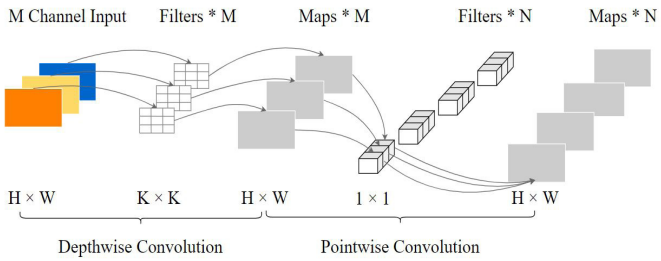


Fig. 2. Depthwise separable convolution.

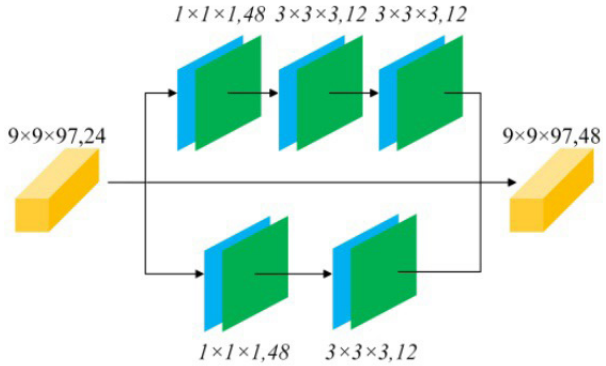


Fig. 3. 3-D two-way dense layer in [35].

where p represents the predicted probability of a sample in the category, and y, an indicate variable equal to 0 or 1, represents whether the sample is correctly classified. It can be observed that when y is 1, the closer the p is to 1, the smaller the loss and if y is 0, the closer the p is to 0, the smaller the loss, which is in line with the direction of optimization.

To simplify the expression, we introduce a piecewise function F as:

$$F_y(t) = \begin{cases} t, & \text{if } y = 1 \\ 1 - t, & \text{otherwise} \end{cases} \tag{2}$$

and then the standard CEL can be reformulated as:

$$\text{CEL}(p, y) = -\log(F_y(p)). \tag{3}$$

In order to handle data with imbalanced categories, a coefficient $\alpha \in [0, 1]$ is added to the standard cross-entropy loss function to balance the weights of samples of different categories, which results in the following balanced CEL (BCEL):

$$\text{BCEL}(p, \alpha, y) = -F_y(\alpha)\log(F_y(p)). \tag{4}$$

C. LiteDenseNet

In 2020, Li and Duan [35] proposed a lightweight network LiteDenseNet. To obtain receptive fields in different scales, the network uses a 3-D two-way dense layer which was proposed in PeleeNet [37] and DBDA [34]. Its schematic is shown in Fig. 3.

This 3-D two-way dense layer comprises a top layer and a bottom layer. After performing a $1 \times 1 \times 1$ convolution transformation on the input data, the top layer uses a two stacked $3 \times 3 \times 3$ convolution to capture global features and obtains an output with size $(9 \times 9 \times 97, 12)$, and the bottom layer uses a convolution kernel with size $3 \times 3 \times 3$ to capture local features and obtains an output with size $(9 \times 9 \times 97, 12)$. The outputs of the two-way dense layer are used to perform the concatenate operation to generate a 3-D cube with size $(9 \times 9 \times 97, 48)$.

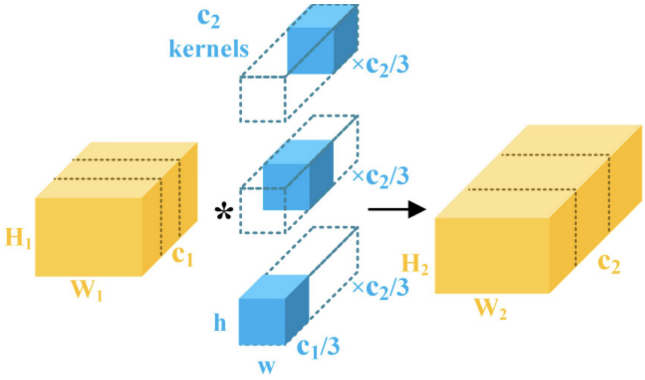


Fig. 4. Group convolution.

However, the 3-D convolutional block in the original 3-D two-way dense layer is a computationally intensive model. In order to reduce the number of parameters and computational cost, LiteDenseNet uses group convolution instead of normal convolution. The group convolution given in [35] is shown in Fig. 4.

Given an input data with size $h \times w \times l$ and c_F channels, the normal convolution layer tends to have the same channel size as the inputs, whereas the group convolution divides the original convolution kernels and the input into 3 groups by channel dimension. As a result, the input is divided into three groups with the size of $h \times w \times l$ and channels $c_F/3$. Three groups are fed into a normal convolution with the $c_F/3$ channels separately. So the channel size is reduced from $h \times w \times l \times c_F$ to $h \times w \times l \times (c_F/3)$. In this way, group convolution significantly reduces the number of parameters and computational cost while maintaining the size of the input and the output.

III. Proposed LiteDepthwiseNet

In this section, we describe the proposed LiteDepthwiseNet in detail.

A. Modified 3-D Depthwise Convolution

LiteDenseNet reduces the parameters and computational cost using group convolution to replace standard convolution. However, group convolution permanently cuts off some connections between channels, which results in reducing the classification performance to some extent. Moreover, for HSI classification, there is still room to reduce the parameters and computational cost. So, we consider replacing group convolution with depthwise separable convolution, in which pointwise convolution can connect all the channels. At the same time, for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

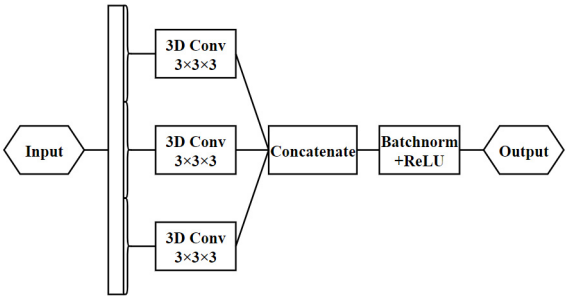


Fig. 5. Group convolution with 3 groups.

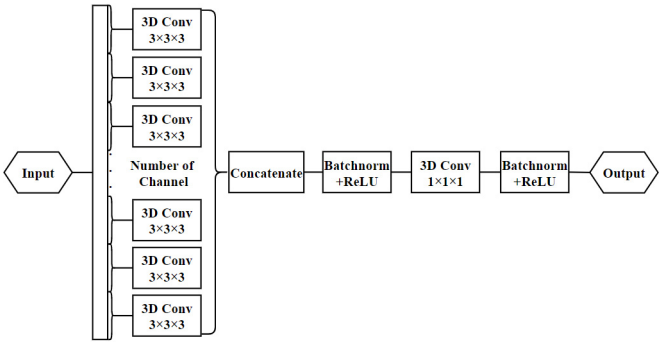


Fig. 6. 3-D depthwise convolution.

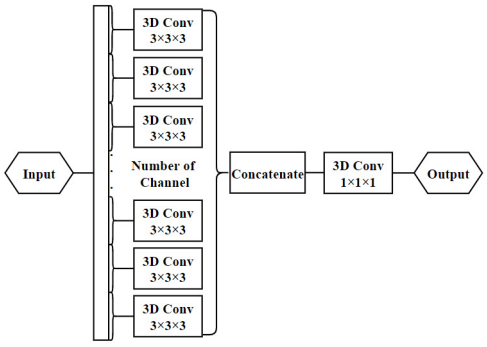


Fig. 7. Modified 3-D depthwise convolution.

HSIs, there are too many channels, so 2-D depthwise separable convolution will still have too many parameters. A good choice is to use 3-D depthwise convolution originally proposed in [38] for RGB images. At this point, a new problem appears. The limited number of training samples for HSIs always causes overfitting for the model with high complexity. So, we propose a modified 3-D depthwise convolution for HSI classification. The major adjustment is that the middle activation layer and normalization layer of the original 3-D depthwise convolution network are stripped, which reduces the complexity of the model and is likely to improve the overfitting phenomenon. Moreover, the number of parameters and the computational consumption are greatly reduced. Structural comparison diagrams of group convolution, 3-D depthwise convolution, and modified 3-D depthwise convolution are shown in Figs. 5–7, respectively.

B. Focal Loss

Since most HSIs have imbalanced data categories [5], using the standard CEL function for training will lead to low classification accuracy for the small sample categories. The BCEL function is commonly used to deal with this situation. However, the BCEL function is also likely to focus on the easy-to-classify samples. In order to improve the classification performance, the model should focus on both small-category samples and difficult-to-classify samples. For this purpose, an FL is proposed in [39] and expressed as

$$\mathrm{FL}(p, \alpha, \gamma, y) = -F_y(\alpha)(1 - F_y(p))^{\gamma} \log(F_y(p)) \quad (5)$$

where $\alpha \in [0, 1]$, $\gamma \geq 0$.

From formula (5), it is easy to know the following three points. First, similar to the BCEL, parameter α is introduced to reflect the imbalanced problem of different categories of samples, which helps to improve the accuracy of the final model. Secondly, the modulating factor $(1 - F_y(p))^{\gamma}$ is added to adjust the weights of easy-to-classify samples and difficult-to-classify samples. When a sample is misclassified and $F_y(p)$ is small, then $(1 - F_y(p))^{\gamma}$ is close to 1, and the loss is not affected. However, if $F_y(p) \to 1$, which indicates that the classification prediction of the sample is better, $(1 - F_y(p))^{\gamma}$ is 0, and the loss is down-weighted. Lastly, the focusing parameter γ smoothly adjusts the down-weighted rate of easy-to-classify samples. When $\gamma = 0$, the FL is equivalent to the BCEL and with the increase of γ, the effect of the modulating factor also increases. Of these, α can be set by category frequency, or it can be obtained as a hyperparameter for cross-validation.

C. Framework of the LiteDepthwiseNet

The overall framework of the proposed LiteDepthwiseNet and the specific parameters in an example is shown in Fig. 8 and Table I, respectively.

Specifically, in our network framework, batch normalization (BN) and ReLU activation (ReLU) are added after each point convolution and ordinary convolution. For the sake of brevity, it will not be specified in the following description. When an input, a 3-D cube of size $(9 \times 9 \times 200, 1)$, is fed into a 3-D CNN layer with size $(1 \times 1 \times 7, 24)$, we get an output with size of $(9 \times 9 \times 97, 24)$. Next we put the output into two branches respectively. The right branch (Branch 1) performs group convolution on the input and divides the input into three groups, where each group is a 3-D cube with the size of $(9 \times 9 \times 97, 8)$. Then, we put them into a 3-D CNN layer with the size of $(1 \times 1 \times 1, 16)$. Until now, we have retained the original design of LiteDenseNet, because modifying the first and second layers to 3-D depthwise convolution increases the number of parameters. After this, the output results of the three 3-D CNNs are stacked by channel dimension, and we obtain the output with size of $(9 \times 9 \times 97, 48)$. The subsequent modules are all composed of 3-D depthwise convolution, which divides $(9 \times 9 \times 97, 48)$ into 48 independent 3-D blocks by channel dimension, and uses 48 3-D CNN filters with the size of $(3 \times 3 \times 3, 1)$ for one-to-one convolution operation, and the size of output is $(9 \times 9 \times 97, 48)$. We feed this into

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

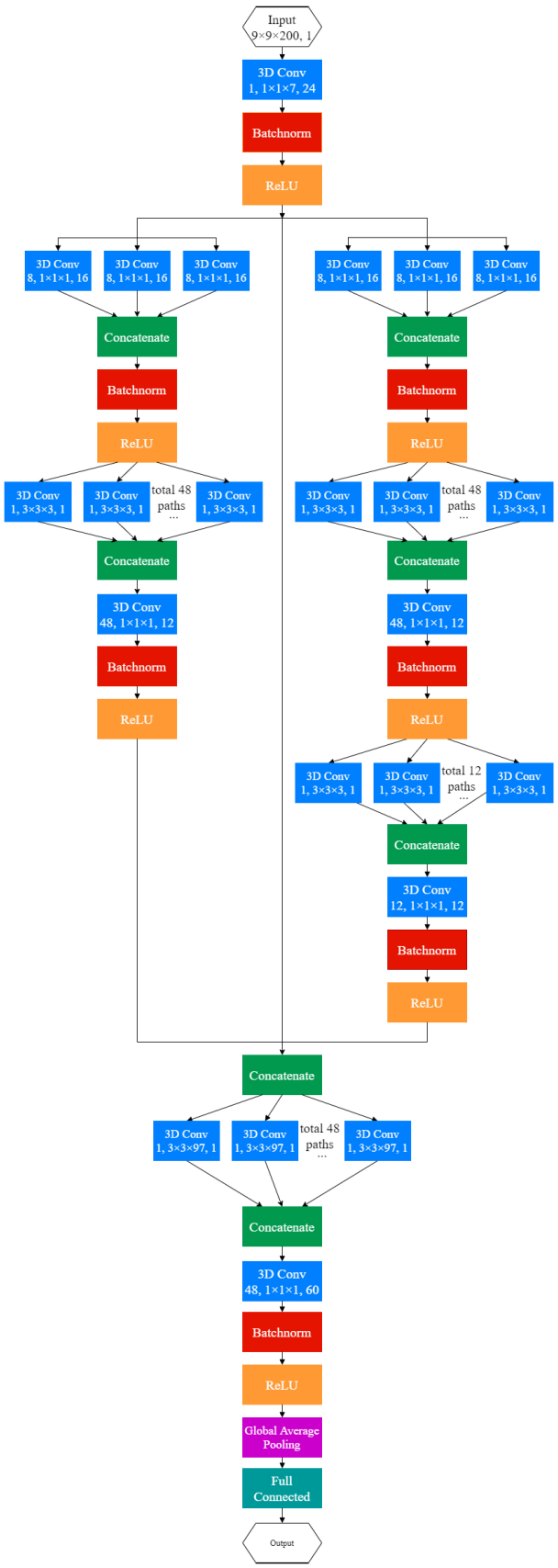


Fig. 8. LiteDepthwiseNet architecture.

the 3-D CNN with the size of $(1 \times 1 \times 1, 12)$ for a point-by-point linear combination, which makes up for the defect of group convolution. Then, we continue to feed a 3-D depthwise convolution, where the number of input and output channels are both 12. At this point, the design of the right branch is completed. The left branch (Branch 2) feeds the input obtained by the group convolution into a 3-D depthwise convolution. Finally, we contact the outputs of the three channels, that is, the input that has not entered any branch with the size of $(9 \times 9 \times 97, 24)$, the output of the right branch with the size of $(9 \times 9 \times 97, 12)$, and the output of the left branch with the size of $(9 \times 9 \times 97, 12)$, to obtain the output with the size of $(9 \times 9 \times 97, 48)$, which is fed into the last 3-D depthwise convolution, and the final result is obtained through a global average pooling layer and a fully connected layer.

TABLE I
IMPLEMENTATION DETAILS ON AN EXAMPLE OF LITEDEPTHWISENET

Layer name		Kernel Size	Group	Output Size
Input		-	-	$(9 \times 9 \times 200, 1)$
3D-CNN+BN+ReLU		$(1 \times 1 \times 7)$	1	$(9 \times 9 \times 97, 24)$
Branch 1	3D-CNN+Concatenate +BN+ReLU	$(1 \times 1 \times 1)$	3	$(9 \times 9 \times 97, 48)$
	3D-CNN+Concatenate	$(3 \times 3 \times 3)$	48	$(9 \times 9 \times 97, 48)$
	3D-CNN+BN+ReLU	$(1 \times 1 \times 1)$	1	$(9 \times 9 \times 97, 12)$
Branch 2	3D-CNN+Concatenate +BN+ReLU	$(1 \times 1 \times 1)$	3	$(9 \times 9 \times 97, 48)$
	3D-CNN+Concatenate	$(3 \times 3 \times 3)$	48	$(9 \times 9 \times 97, 48)$
	3D-CNN+BN+ReLU	$(1 \times 1 \times 1)$	1	$(9 \times 9 \times 97, 12)$
	3D-CNN+Concatenate	$(3 \times 3 \times 3)$	12	$(9 \times 9 \times 97, 12)$
	3D-CNN+BN+ReLU	$(1 \times 1 \times 1)$	1	$(9 \times 9 \times 97, 12)$
Concatenate		-	-	$(9 \times 9 \times 97, 48)$
3D-CNN+Concatenate		$(3 \times 3 \times 97)$	48	$(9 \times 9 \times 1, 48)$
3D-CNN+BN+ReLU		$(1 \times 1 \times 1)$	1	$(9 \times 9 \times 1, 60)$
Global Average Pooling		-	-	(1×60)
Full Connected		-	-	(1×16)

D. Comparing Calculation Cost and Number of Parameters

Under certain conditions, compared with group convolution, our modified 3-D depthwise convolution reduces more parameters and has lower computation cost. For example, taking $(h \times w \times l)$, c_F as the size and the number of channels of the input, $(h_F \times w_F \times l_F)$ as the convolution kernel size, c_G as the size and the number of channels of the output, the number of parameters is given by

$$(h \times w \times l \times (c_F/3)) \times (c_G/3) \times 3$$

and the amount of the computation of group convolution is calculated as

$$(h \times w \times l \times (c_F/3) \times h_F \times w_F \times l_F \times (c_G/3)) \times 3$$

whereas the number of parameters of the 3-D depthwise convolution is as follows,

$$h \times w \times l \times c_F \times 1 + 1 \times 1 \times 1 \times c_F \times c_G$$

and the amount of the computation of 3-D depthwise convolution is

$$h \cdot w \cdot l \cdot c_F \cdot h_F \cdot w_F \cdot l_F + c_F \cdot c_G \cdot h_F \cdot w_F \cdot l_F.$$

For some special cases, such as $1 \times 1 \times 1$ convolution kernel, the 3-D depthwise convolution increases the number of parameters and the computation cost.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF EIGHT ALGORITHMS

Network	Input size	FLOPs	Parameters
ChenEtAlNet	$25 \times 25 \times 200$	6.030G	1.120M
HybridSN	$25 \times 25 \times 200$	2.483G	8.256M
HamidaEtAlNet	$25 \times 25 \times 200$	1.148G	6.452M
LeeEtAlNet	$25 \times 25 \times 200$	**243.930M**	390.288k
AINet	$25 \times 25 \times 200$	381.499M	659.230k
DBDA	$25 \times 25 \times 200$	815.981M	382.326k
LiteDenseNet	$25 \times 25 \times 200$	1.316G	852.309k
LiteDepthwiseNet	$25 \times 25 \times 200$	369.331M	**51.616k**

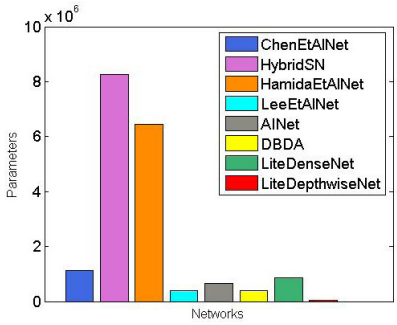


Fig. 9. Parameters of each algorithm.

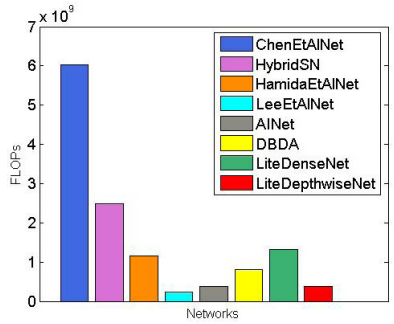


Fig. 10. FLOPs of each algorithm.

Table II, Figs. 9 and 10 amount compare the results of the number of parameters and of the computation for our proposed LiteDepthwiseNet and five other DL algorithms when the input size is $25 \times 25 \times 200$. Of these, for the HybridSN method, PCA is required for preprocessing. For the convenience of comparison, this operation is not used to calculate the computation overhead.

We can see that the number of parameters of the proposed LiteDepthwiseNet is much smaller than that of the other five algorithms. In particular, it is 93.94% less than the LiteDenseNet. Moreover, the computation cost of LiteDepthwiseNet is 71.91% less than that of the LiteDenseNet. Note that the LeeEtAlNet has a minimal computation cost, but later experiment results show that its classification accuracy is far lower than ours. Note that the proposed algorithm only needs to use float 16 for training and inference to the state-of-the-art performance, which further reduces the overhead of computing resources.

IV. EXPERIMENT RESULTS

In this section, we perform experiments on five widely used hyperspectral data sets to compare the performance of the proposed LiteDepthwiseNet with eight other methods, namely, LiteDenseNet [35], DBDA [34], AINet [33], LeeEtAlNet [27], HamidaEtAlNet [29], HybridSN [30], ChenEtAlNet [28] and the classic support vector machine (SVM) [7]. Except for LiteDenseNet, the other algorithms are tested using their publicly recognized open source codes or the codes provided by the authors. Since LiteDenseNet encapsulates its code, we use our own reproduced code with balanced cross-entropy loss for the experiments. All the experiments are implemented on Ubuntu 18.04.3 LTS, CPU: i9 9900k, GPU: 2 × 2080ti, RAM: 64 GB, GPU memory: 22 GB, Python: 3.6.1, Torch: 1.3.1. Three quantitative indicators, overall accuracy (OA), average accuracy (AA) and κ coefficient (Kappa), are used to evaluate the performance of each method.

A. Data Description

The following five open HSI data sets are used for the experiments in this article. They can be obtained from http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes and https://hyperspectral.ee.uh.edu/?page_id=459.

1) Indian Pines (IP): This data set was acquired by the airborne visible infrared imaging spectrometer (AVIRIS) sensor in north-western Indiana. The size of the scene is $145 \times 145 \times 200$ with a spatial resolution 20 m per pixel and spectral coverage ranging from 0.4 to 2.5 μm. The ground truth (GT) available is divided into 16 classes of vegetation.

2) University of Pavia (UP): This data set captures an urban area surrounding the UP, Italy, which was collected by the reflective optics imaging spectrometer (ROSIS-03) sensor in Northern Italy in 2001. The size of the scene is $610 \times 340 \times 103$ with a spatial resolution of 1.3 m per pixel and spectral coverage ranging from 0.43 to 0.86 μm. The GT is designated into 9 urban land-cover classes.

3) Pavia Center (PC): This data set is also acquired by the reflective optics imaging spectrometer (ROSIS-03) sensor. The size of the scene is $1096 \times 715 \times 102$ with a spatial resolution 1.3 m per pixel and spectral coverage ranging from 0.43 to 0.86 μm. The GT available is divided into 9 classes.

4) Houston (HS): This data set is acquired by an airborne sensor over the area of the University of Houston campus and the neighboring urban area. The size of the scene is $349 \times 1905 \times 144$ with a spatial resolution 2.5 m per pixel and spectral coverage ranging from 0.38 to 1.05 μm. The GT available is divided into 15 classes.

5) Botswana (BS): This data set is acquired by the NASA EO-1 satellite over the Okavango Delta. The size of the scene is $1476 \times 256 \times 145$ with a spatial resolution 30 m per pixel and spectral coverage ranging from 0.4 to 2.5 μm. The GT available is divided into 14 classes.

In the follow-up experiments, the UP, PC, HS, and BS data sets are divided into training, verification, and testing sets, and the ratio of training, verification, and testing within each class is the same. Here, we choose 0.5% of the UP data set for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III

SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION AND TESTING FOR THE IP DATA SET

No.	Class	Total number	Training	Validation	Testing
1	Alfal	46	5	-	41
2	Corn-notill	1428	71	-	1357
3	Corn-mintill	830	41	-	789
4	Corn	237	12	-	225
5	Grass-pasture	483	24	-	459
6	Grass-trees	730	37	-	693
7	Grass-pasture-mowed	28	5	-	23
8	Hay-windrowed	478	24	-	454
9	Oats	20	5	-	15
10	Soybean-notill	972	49	-	923
11	Soybean-mintill	2455	109	-	2346
12	Soybean-clean	593	30	-	563
13	Wheat	205	12	-	193
14	Woods	1265	63	-	1202
15	Buildings-Grass-Trees	386	19	-	367
16	Stone-Steel-Towers	93	6	-	87
	Total	10249	512	-	9737

TABLE IV

SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION AND TESTING FOR THE UP DATA SET

No.	Class	Total number	Training	Validation	Testing
1	Asphalt	6631	33	33	6565
2	Meadows	18649	93	93	18463
3	Gravel	2099	10	11	2078
4	Trees	3064	15	16	3033
5	Painted metal sheets	1345	7	6	1332
6	Bare Soil	5029	25	25	4979
7	Bitumen	1330	7	6	1317
8	Self-Blocking Bricks	3682	19	19	3644
9	Shadows	947	5	5	937
	Total	42776	214	214	42348

TABLE V

SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION AND TESTING FOR THE PC DATA SET

No.	Class	Total number	Training	Validation	Testing
1	Water	65971	65	65	65841
2	Trees	7598	7	7	7584
3	Meadow	3090	3	3	3084
4	Brick	2685	3	3	2679
5	Soil	6584	6	6	6572
6	Asphalt	9248	9	9	9230
7	Bitumen	7287	7	7	7273
8	Tile	42826	42	42	42742
9	Shadow	2863	3	3	2857
	Total	148152	145	145	147862

TABLE VI

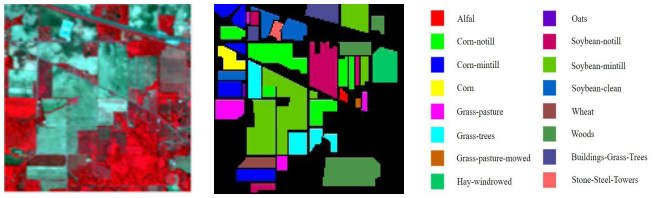
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION AND TESTING FOR THE HS DATA SET

No.	Class	Total number	Training	Validation	Testing
1	Healthy grass	1251	25	25	1201
2	Stressed grass	1254	25	25	1204
3	Synthetic grass	697	14	14	669
4	Trees	1244	25	25	1194
5	Soil	1242	25	25	1192
6	Water	325	7	6	312
7	Residential	1268	25	26	1217
8	Commercial	1244	25	25	1194
9	Road	1252	25	25	1202
10	Highway	1227	24	25	1178
11	Railway	1235	24	25	1186
12	Parking Lot 1	1233	24	25	1184
13	Parking Lot 2	469	10	9	450
14	Tennis Court	428	9	8	411
15	Running Track	660	13	13	634
	Total	15029	300	301	14428

TABLE VII

SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION AND TESTING FOR THE BS DATA SET

No.	Class	Total number	Training	Validation	Testing
1	Water	270	4	4	262
2	Hippo grass	101	2	1	98
3	Floodplain grasses1	251	4	4	243
4	Floodplain grasses2	215	3	4	208
5	Reeds1	269	4	4	261
6	Riparian	269	4	4	261
7	Fierscar2	259	4	4	251
8	Island interior	203	3	3	197
9	Acacia woodlands	314	4	5	305
10	Acacia shrublands	248	3	4	241
11	Acacia grasslands	305	4	5	296
12	Short mopane	181	3	2	176
13	Mixed mopane	268	4	4	260
14	Exposed soils	95	2	1	92
	Total	3248	48	49	3151



(a) (b)

Fig. 11. IP data set. (a) Three-band false color composite. (b) Ground reference data and color code.

training, 0.1% of the PC data set for training, 2% of the HS data set for training, 1.5% of the BS data set for training, and the number of samples in the training set and verification set of the UP, PC, HS, and BS data sets are the same. But for the IP data set, the training ratio is too low, which will lead to the absence of training samples in some categories. To ensure that there are at least 5 samples in each category, we use 5% of the IP data set for training and do not set a verification set. The detailed division information of the five data sets is listed in Tables III–VII, respectively. The false color composite of different data sets and the corresponding ground reference data are shown in Figs. 11–15. Among them, the false color composite of BS is from [34] and that of HS is from [5].

B. Classification Maps and Categorized Results

1) Classification Maps and Categorized Results for the IP Data Set: The results of using different algorithms to classify the IP data set are shown in Table VIII, and the classification maps of the different algorithms and the GT categories are shown in Fig. 16. The main characteristic of the IP data set is that the amount of data is small and the data distribution is imbalanced. In particular, the amount of data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
CATEGORIZED RESULTS FOR THE IP DATA SET USING 5% TRAINING SAMPLES

Class	SVM	ChenEtAlNet	HybridSN	HamidaEtAlNet	LeeEtAlNet	AINet	DBDA	LiteDenseNet	LiteDepthwiseNet
1	9.09	31.82	34.09	0.00	0.00	20.45	89.52	**100.00**	**100.00**
2	73.03	34.83	89.17	0.00	27.93	67.06	93.07	91.97	**97.42**
3	55.00	48.81	78.83	7.10	0.00	55.39	88.40	**95.31**	94.68
4	32.00	45.24	61.78	0.00	36.89	56.89	96.39	**98.67**	98.22
5	89.11	46.72	79.52	0.00	0.00	66.23	**98.73**	88.89	88.89
6	95.96	92.58	98.56	96.54	91.92	88.89	**99.57**	99.28	99.28
7	59.26	32.00	11.11	0.00	0.00	11.11	67.00	**100.00**	**100.00**
8	99.56	86.04	96.26	92.95	95.37	99.34	**99.91**	99.78	99.78
9	15.79	17.65	**100.00**	0.00	0.00	10.53	78.82	**100.00**	86.67
10	62.19	72.46	79.52	0.00	0.00	65.66	**92.87**	92.20	90.68
11	75.17	81.49	94.85	94.60	84.18	69.38	94.92	96.12	**96.59**
12	49.02	29.54	82.77	0.00	0.00	48.13	94.42	**95.74**	**95.74**
13	93.33	84.10	79.49	1.54	0.00	84.10	**99.68**	96.89	95.34
14	93.18	78.39	98.17	99.42	96.42	94.09	98.16	96.34	**99.58**
15	52.59	51.39	68.94	0.27	1.91	46.59	85.76	**97.55**	95.37
16	94.32	6.90	50.00	0.00	0.00	94.32	87.17	98.85	**100.00**
OA ×100	74.22	66.00	87.68	46.75	47.87	70.90	93.94	95.35	**96.29**
AA ×100	61.68	52.52	75.19	23.08	25.57	61.14	91.52	**96.72**	96.14
Kappa ×100	70.50	59.39	85.89	35.60	37.60	66.86	93.09	94.70	**95.77**
FLOPs		6.030G	2.483G	1.148G	**243.930M**	381.499M	815.981M	1.316G	369.331M
Parameters		1.120M	8.256M	6.452M	390.288k	659.230k	382.326k	852.309k	**51.616k**

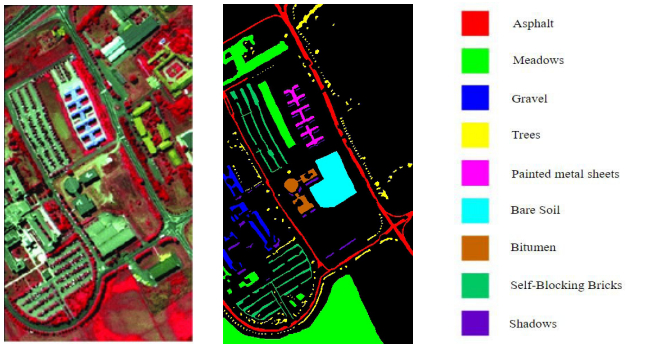


Fig. 12. UP data set. (a) Three-band false color composite. (b) Ground reference data and color code.

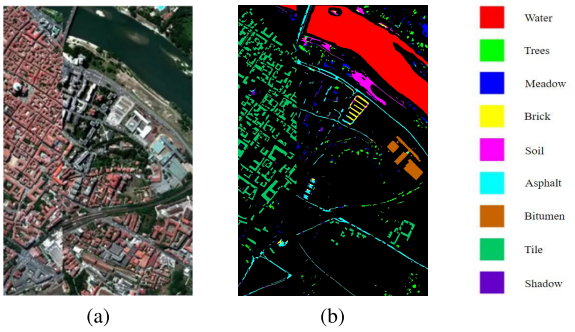


Fig. 13. PC data set. (a) Three-band false color composite. (b) Ground reference data and color code.

in categories 1, 7, 9, and 16 is less than 100, far less than that in other categories. Using the training data in Table XIII, the proposed algorithm performs best in OA and Kappa compared with the other six algorithms. On most categories, LiteDepthwiseNet achieves similar results to LiteDenseNet with 6.06% parameters and 28.09% FLOPs. In the small sample categories, compared with other four large networks, the two lightweight networks show good classification properties. Among them, for the 9th category, our proposed network performs poorly, while HybridSN and LiteDenseNet showed

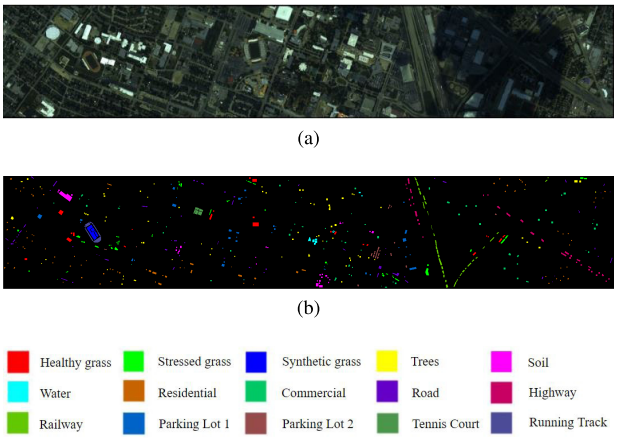


Fig. 14. HS data set. (a) Three-band false color composite. (b) Ground reference data and color code.

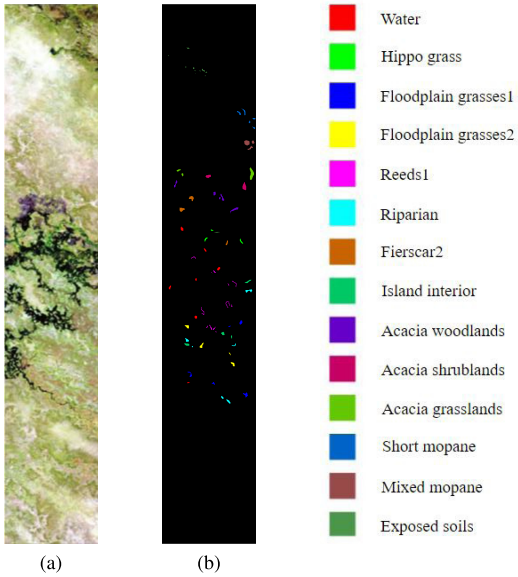


Fig. 15. BS data set. (a) Three-band false color composite. (b) Ground reference data and color code.

100% accuracy. It is worth noting that the overall performance of HybridSN is not satisfactory. LeeEtAlNet, the only one that is less computationally expensive than our proposed method,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

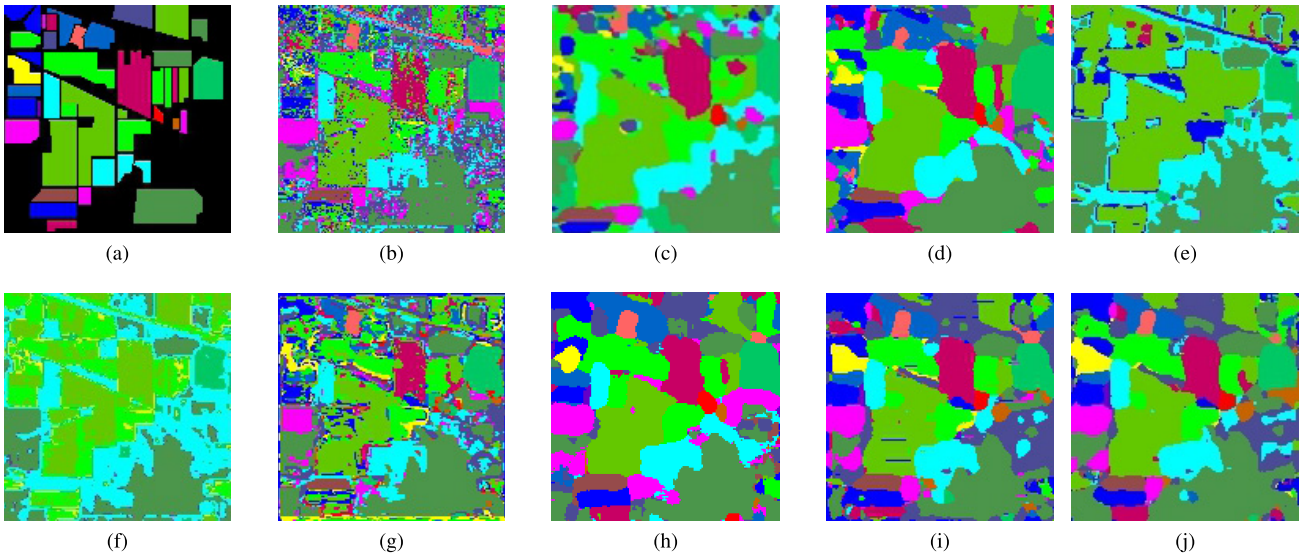


Fig. 16. Classification maps obtained on the extracted features from IP data set with 5% of training samples. (a) GT. (b) SVM. (c) ChenEtAlNet. (d) HybridSN. (e) HamidaEtAlNet. (f) LeeEtAlNet. (g) AINet. (h) DBDA. (i) LiteDenseNet. (j) LiteDepthwiseNet.

TABLE IX

CATEGORIZED RESULTS FOR THE UP DATA SET USING 0.5% TRAINING SAMPLES

Class	SVM	ChenEtAlNet	HybridSN	HamidaEtAlNet	LeeEtAlNet	AINet	DBDA	LiteDenseNet	LiteDepthwiseNet
1	90.77	94.41	92.73	96.60	87.24	74.24	96.66	93.15	**97.15**
2	98.23	68.13	98.21	90.25	98.42	89.11	99.44	99.21	**99.77**
3	51.97	0.00	78.36	2.65	7.51	18.82	**89.71**	84.84	78.10
4	88.20	**98.55**	82.06	76.26	69.04	61.92	97.35	95.91	96.21
5	98.72	99.93	**100.00**	96.85	99.92	92.04	99.26	99.25	99.40
6	62.56	0.58	97.78	25.29	2.13	40.35	97.75	**98.35**	96.77
7	61.12	0.00	41.50	0.23	0.00	61.73	93.39	88.84	**96.20**
8	89.19	90.51	44.24	76.60	0.03	67.85	82.61	95.86	**97.97**
9	**100.00**	95.52	73.67	**100.00**	99.89	83.88	97.59	98.51	98.72
OA ×100	88.11	64.50	88.28	74.75	67.35	72.97	96.34	96.60	**97.39**
AA ×100	73.95	54.76	78.73	56.47	46.42	65.55	94.86	94.88	**95.59**
Kappa ×100	83.90	53.90	84.27	65.40	53.50	63.76	95.14	95.50	**96.54**
FLOPs		1.953G	1.208G	591.067M	**165.766M**	196.519M	815.981M	664.921M	187.531M
Parameters		1.068M	6.467M	1.975M	265.225k	657.945k	382.326k	437.157k	**30.453k**

TABLE X

CATEGORIZED RESULTS FOR THE PC DATA SET USING 0.1% TRAINING SAMPLES

Class	SVM	ChenEtAlNet	HybridSN	HamidaEtAlNet	LeeEtAlNet	AINet	DBDA	LiteDenseNet	LiteDepthwiseNet
1	99.96	93.98	**99.98**	99.10	99.76	98.95	99.89	99.83	99.80
2	91.15	85.56	78.80	96.35	**99.91**	95.23	93.77	94.14	91.82
3	**93.30**	21.37	59.73	67.47	10.57	43.00	75.24	84.53	93.09
4	59.51	22.05	31.77	39.26	32.90	31.46	70.23	71.75	**97.50**
5	96.03	80.98	70.66	94.34	83.13	36.16	**97.70**	95.80	97.52
6	96.92	67.16	75.13	97.91	**99.03**	93.85	92.97	97.27	96.61
7	83.35	92.21	85.98	79.59	81.42	68.06	100.00	**91.97**	90.87
8	99.38	93.49	99.96	99.22	99.95	88.12	99.57	99.37	**99.98**
9	**99.93**	58.24	6.54	97.79	96.28	70.21	99.06	86.24	85.33
OA	97.28	87.46	91.46	95.98	94.99	88.05	97.62	97.59	**98.24**
AA	81.95	61.30	67.62	77.10	70.29	69.45	92.05	91.21	**94.73**
Kappa ×100	96.20	82.60	87.70	94.30	92.90	83.07	96.62	96.58	**97.51**
FLOPs		1.953G	1.195G	577.268M	**164.966M**	193.775M	815.981M	651.353M	183.743M
Parameters		1.068M	6.448M	1.837M	263.945k	657.945k	382.326k	428.517k	**30.021k**

is already unacceptable in terms of accuracy. DBDA has a very good performance on the IP data set, and for AINet, there is a big gap between this algorithm and our algorithm in accuracy.

2) Classification Maps and Categorized Results for the UP Data Set: The results of using different algorithms to classify the UP data set are shown in Table IX, and the classification maps of different algorithms and the ground truth are shown in Fig. 17. The main characteristics of this data are a large amount of data and the balanced distribution of categories. In the case of using only 0.5% of the training samples, our proposed method achieves state-of-the-art performance on OA, AA, and Kappa. It can be clearly seen that some large networks that performed well in the past do not perform well at a small training rate, such as HybridSN, which only achieves an OA of 87.68%. Our proposed LiteDepthwiseNet generates the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

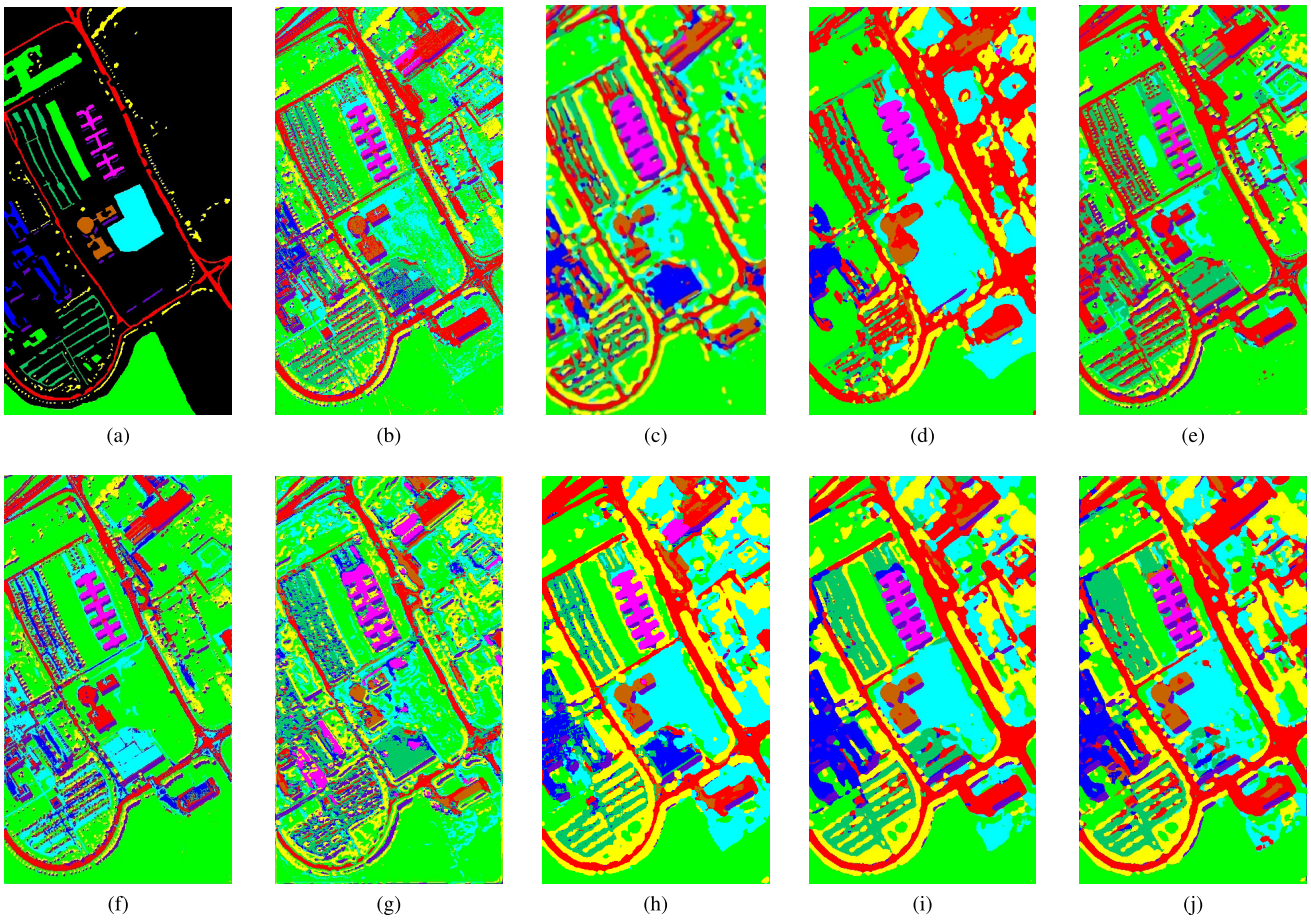


Fig. 17. Classification maps obtained on the extracted features from UP data set with 0.5% of training samples. (a) GT. (b) SVM. (c) ChenEtAlNet. (d) HybridSN. (e) HamidaEtAlNet. (f) LeeEtAlNet. (g) AINet. (h) DBDA. (i) LiteDenseNet. (j) LiteDepthwiseNet.

TABLE XI
CATEGORIZED RESULTS FOR THE HS DATA SET USING 2% TRAINING SAMPLES

Class	SVM	ChenEtAlNet	HybridSN	HamidaEtAlNet	LeeEtAlNet	AINet	DBDA	LiteDenseNet	LiteDepthwiseNet
1	90.17	8.41	97.39	72.27	92.73	97.34	99.37	**100.00**	99.83
2	**98.75**	88.46	94.14	67.77	93.68	86.30	93.91	87.21	97.84
3	99.40	96.56	97.07	96.71	90.68	87.00	**100.00**	**100.00**	**100.00**
4	93.05	29.65	89.17	92.80	93.74	87.02	97.36	92.21	93.55
5	98.41	39.26	**100.00**	88.00	85.14	89.18	**100.00**	99.50	**100.00**
6	84.29	36.86	82.13	71.15	83.70	70.51	**97.49**	96.79	87.82
7	85.37	68.12	69.59	86.52	53.85	52.75	86.63	95.65	**96.14**
8	57.96	54.10	89.91	45.81	62.39	57.29	**100.00**	69.68	74.96
9	75.04	61.81	64.87	69.80	71.68	79.45	86.94	**93.93**	93.26
10	81.58	91.00	**99.58**	65.53	38.25	54.07	77.48	92.53	92.11
11	67.79	54.38	88.84	62.90	60.95	66.78	**94.48**	85.50	94.27
12	58.36	56.08	84.27	11.32	67.09	70.19	85.80	**96.88**	92.31
13	10.11	36.89	50.43	1.56	5.11	79.33	94.09	89.78	**94.67**
14	92.21	89.78	95.47	90.51	86.39	85.40	**100.00**	99.76	97.81
15	98.26	97.63	**100.00**	99.05	91.52	84.23	96.79	**100.00**	**100.00**
OA $\times 100$	80.54	58.93	87.63	67.98	43.63	75.51	92.55	92.41	**94.07**
AA $\times 100$	74.42	56.81	86.86	63.86	67.31	76.46	94.02	93.29	**94.30**
Kappa $\times 100$	78.90	55.70	86.62	65.30	41.01	73.52	91.94	91.80	**93.59**
FLOPs		3.676G	1.747G	823.783M	**199.049M**	273.295M	815.981M	936.271M	263.281M
Parameters		1.090M	7.223M	4.430M	318.479k	659.487k	382.326k	610.323k	**39.459k**

best classification results with minimal parameters. However, the accuracy of DBDA on the UP data set is slightly lower than our proposed method, but its computational complexity is much higher than our proposed method. It is still difficult for AINet to achieve a reliable accuracy with such a small number of training samples.

3) Classification Maps and Categorized Results for the PC Data Set: Table X and Fig. 18 show the classification results and maps of different algorithms on the PC data set, respectively. The overall sample size of the PC data set is large and basically balanced. Among them, the first and eighth categories are the two categories with the largest number of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

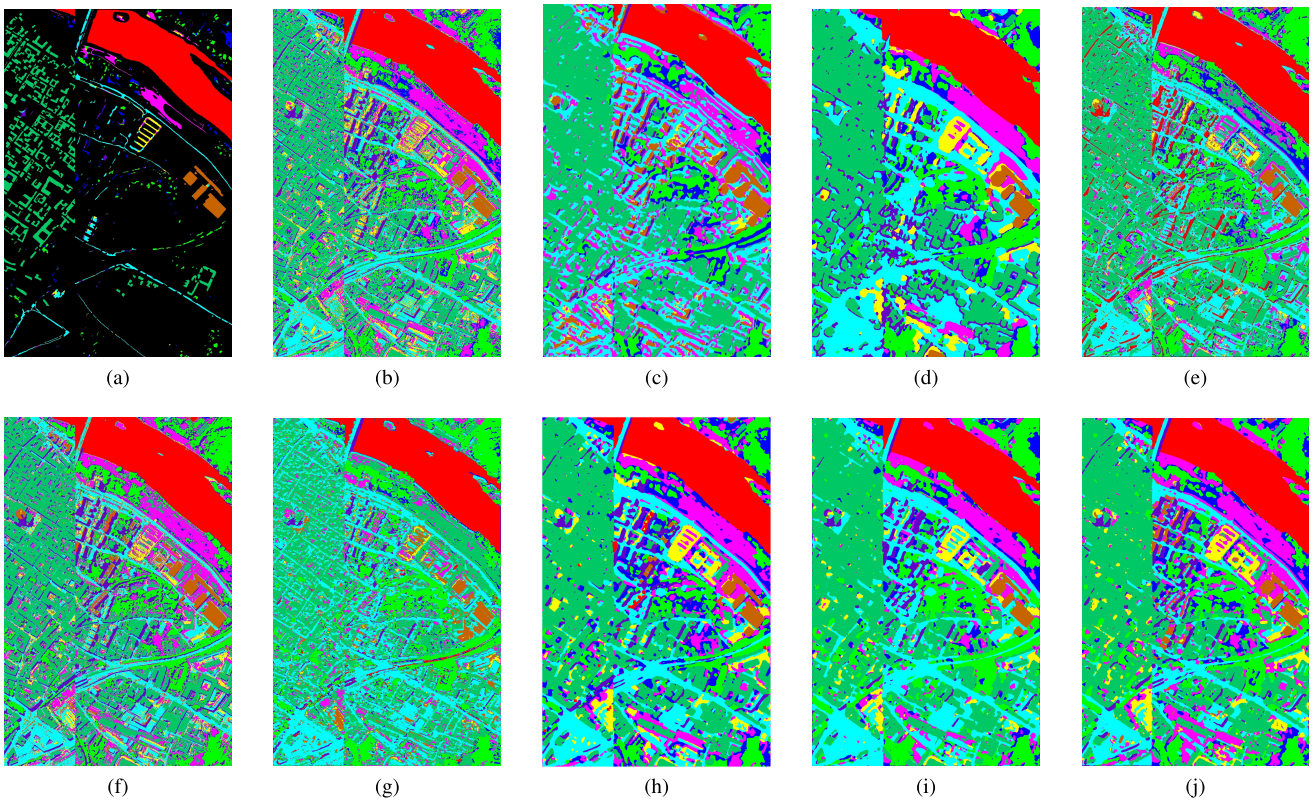


Fig. 18. Classification maps obtained on the extracted features from PC data set with 0.1% of training samples. (a) GT. (b) SVM. (c) ChenEtAlNet. (d) HybridSN. (e) HamidaEtAlNet. (f) LeeEtAlNet. (g) AINet. (h) DBDA. (i) LiteDenseNet. (j) LiteDepthwiseNet.

TABLE XII
CATEGORIZED RESULTS FOR THE BS DATA SET USING 1.5% TRAINING SAMPLES

Class	SVM	ChenEtAlNet	HybridSN	HamidaEtAlNet	LeeEtAlNet	AINet	DBDA	LiteDenseNet	LiteDepthwiseNet
1	**100.00**	0.00	80.08	**100.00**	**100.00**	93.51	99.62	**100.00**	98.85
2	88.78	0.00	38.00	62.24	0.00	79.59	97.27	**100.00**	**100.00**
3	90.53	0.00	74.49	95.06	6.17	86.42	96.27	**100.00**	**100.00**
4	86.54	0.00	82.55	35.10	0.00	68.75	91.32	**100.00**	**100.00**
5	54.02	0.00	55.47	45.98	13.41	68.97	92.06	**94.64**	90.80
6	54.41	0.00	75.47	62.84	0.00	48.66	95.58	62.07	**84.29**
7	96.41	0.00	98.43	89.24	94.02	54.98	**100.00**	**100.00**	**100.00**
8	72.08	0.00	64.00	97.46	65.48	52.79	95.16	**100.00**	**100.00**
9	83.93	**100.00**	96.44	98.69	79.02	62.30	88.23	**100.00**	**100.00**
10	83.40	0.00	87.70	95.02	39.42	82.57	87.85	**100.00**	**100.00**
11	87.16	0.00	89.70	0.00	83.11	92.91	**100.00**	**100.00**	**100.00**
12	87.50	0.00	**100.00**	51.14	0.00	96.59	**100.00**	**100.00**	**100.00**
13	87.69	0.00	42.80	54.62	79.62	**100.00**	97.75	**100.00**	**100.00**
14	97.83	0.00	43.62	91.30	0.00	84.78	**100.00**	83.70	91.30
OA ×100	82.61	9.68	76.56	68.96	46.53	76.07	94.69	95.93	**97.59**
AA ×100	78.02	6.67	73.48	65.25	37.35	76.63	95.79	95.74	**97.52**
Kappa ×100	81.10	0.80	74.57	66.40	41.60	74.05	94.25	95.60	**97.39**
FLOPs		3.718G	2.483G	832.781M	**199.769M**	275.030M	815.981M	949.838M	589.003M
Parameters		1.088M	8.256M	4.136M	319.630k	659.230k	382.326k	618.902k	**30.326k**

samples, far exceeding other categories. Therefore, the OA on this data set is easy to achieve a higher score, but the difficulty lies in the classification of the other seven categories with a small sample size. It can be seen from the above Table X, all the methods achieve good performance and the proposed network achieves the highest OA, AA, and Kappa with the smallest amount of parameters, and the amount of calculation is much lower than that of LiteDenceNet. And the LeeEtAlNet still cannot achieve acceptable accuracy on four small sample categories. The accuracy of AINet on the PC data set still has a certain gap compared with the method we proposed. DBDA has reached more than 90% in OA, but the learning of individual categories is relatively poor.

4) Classification Maps and Categorized Results for the HS Data Set: Table XI and Fig. 19 show the classification results and maps of different algorithms on the HS data set, respectively. The HS data set has a balanced distribution of categories and a large amount of data. It can be seen from Table XI that the method proposed in this article uses only 2% of the training data to reach an OA of 94.85%. DBDA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

TABLE XIII
COMPUTING RESOURCES OF EIGHT DL ALGORITHMS

Network	GPU Memory	Inference Runtime
ChenEtAlNet	2272MB	0.89s
HybridSN	4644MB	2.80s
HamidaEtAlNet	2580MB	0.26s
LeeEtAlNet	1248MB	0.51s
AINet	2339MB	0.81s
DBDA	6510MB	0.55s
LiteDenseNet	4136MB	1.54s
LiteDepthwiseNet	2613MB	0.98s

and LiteDenseNet have obtained similar results with more computing costs. LeeEtAlNet gets the worst result on this data. And there is a certain gap between other algorithms and the former three algorithms in accuracy.

5) Classification Maps and Categorized Results for the BS Data Set: Table XII and Fig. 20 show the classification results and maps of different algorithms on the BS data set, respectively. There are not many samples in the BS data set, but our algorithm only uses very small training samples (i.e., at most 4 samples per class), showing excellent classification performance, and can stably reach 100% accuracy in many categories. LiteDenseNet also shows good performance in many categories, but cannot achieve good classification performance on Riparian and Exposed soils, resulting in an OA rate lower than our proposed method. DBDA also has a good classification performance, similar to LiteDensenet, and the performance is more balanced, there is no category with particularly low accuracy. Other algorithms are unable to achieve a good performance on such a small number of training samples and have a large accuracy gap with the proposed method. The BS data set is a representative data set with very sparse and small number of labeled samples. Our algorithm still has the best performance and low computational cost, which proves the robustness of the proposed method in this article.

C. Analysis of Computing Resources

Given an input data with size (25, 25, 200) and mini batch 64, number of channel 1, the computing resources of eight DL algorithms used in this article are detailed in Tabel XIII. For the three algorithms, DBDA, LiteDenseNet, LiteDepthwiseNet, which have a good learning effect on five data sets, our proposed algorithm has obvious advantages in GPU memory consumption and moderate performance in runtime. For other algorithms, such as LeeEtAlNet, although some have smaller GPU memory and some have less runtime, their classification accuracy is far from satisfactory.

D. Effectiveness of Modified 3-D Depthwise Convolution

This part explores the effect of modifying the structure of the original 3-D depthwise convolution. We have carried out comparative experiments on five data sets. Table XIV shows the results for removing the ReLU and BN in the 3-D

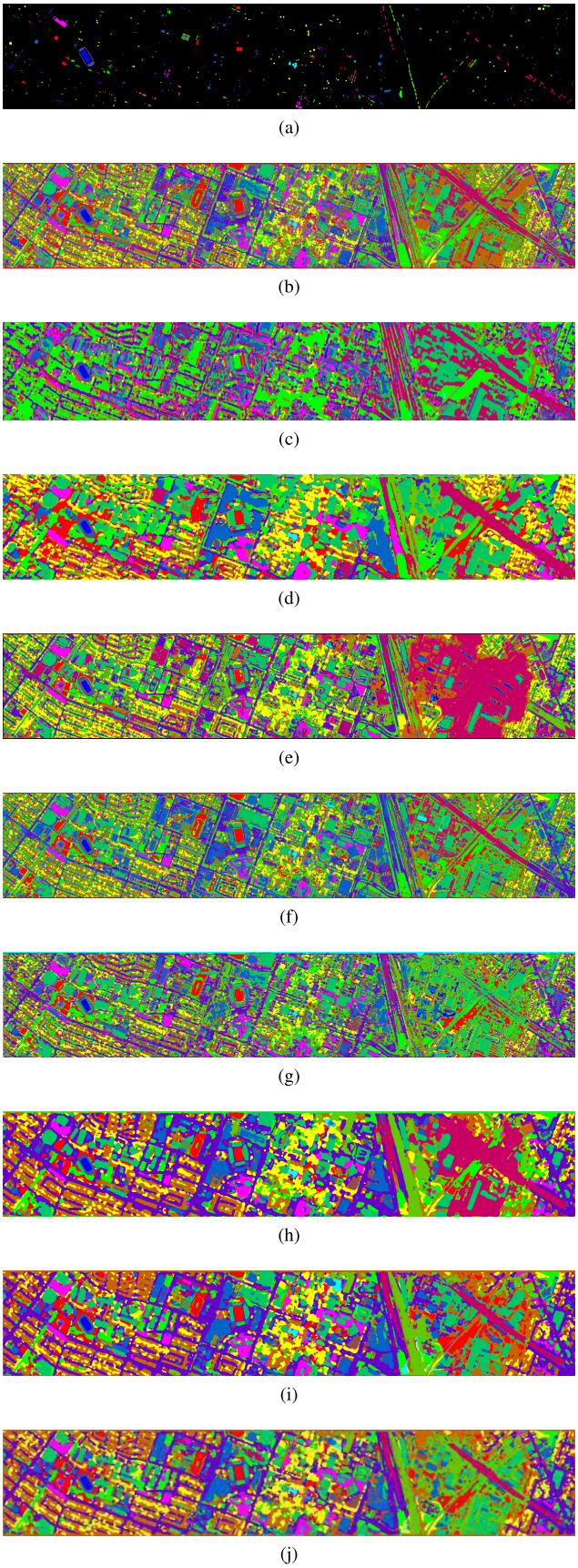


Fig. 19. Classification maps obtained on the extracted features from HS data set with 2% of training samples. (a) GT. (b) SVM. (c) ChenEtAlNet. (d) HybridSN. (e) HamidaEtAlNet. (f) LeeEtAlNet. (g) AINet. (h) DBDA. (i) LiteDenseNet. (j) LiteDepthwiseNet.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

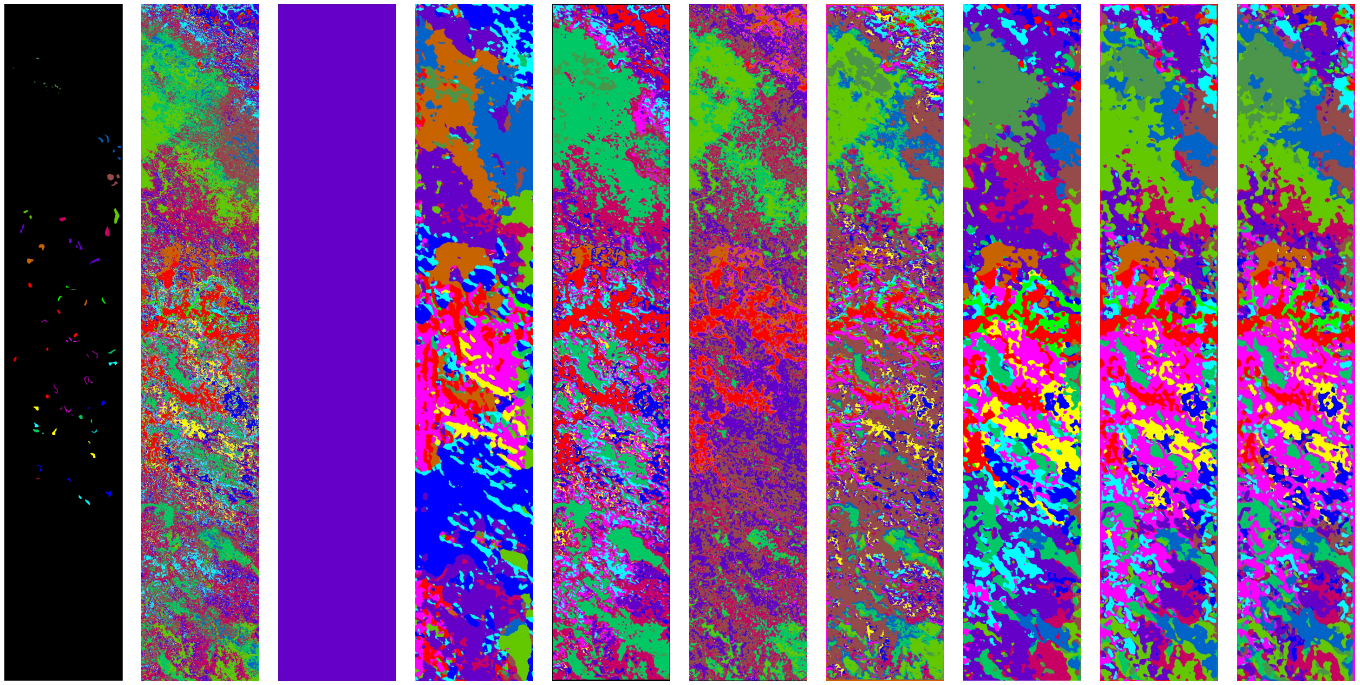


Fig. 20. Classification maps obtained on the extracted features from BS data set with 1.5% of training samples. (a) GT. (b) SVM. (c) ChenEtAlNet. (d) HybridSN. (e) HamidaEtAlNet. (f) LeeEtAlNet. (g) AINet. (h) DBDA. (i) LiteDenseNet. (j) LiteDenseNet.

TABLE XIV
COMPARISON OF 3-D DEPTHWISE CONVOLUTION BEFORE AND AFTER MODIFIED

Dataset	3D Depthwise Convonlution			Modified 3D Depthwise Convonlution		
	OA	AA	Kappa	OA	AA	Kappa
IP	39.23	41.03	33.05	96.29	96.14	95.77
UP	75.24	68.54	66.20	97.39	95.59	96.54
PC	28.15	10.87	16.74	98.24	94.73	97.51
HS	63.17	61.63	60.08	94.07	94.30	93.59
BS	89.30	89.08	88.40	97.59	97.52	97.39

depthwise convolution or not. It can be seen that on each data set, if the ReLU and BN are added between the pointwise convolution and the depthwise convolution, the effect will be very bad. For the original version of 3-D depthwise convolution, too few training samples will cause the model to overfitting, making it perform poorly on the test set. We think that it is because the ReLU and BN in the middle of the separation convolution lead to the high complexity of the model and the too strong fitting ability. Therefore, we removed the ReLU and BN between the pointwise convolution and the depthwise convolution to reduce the model fitting ability to avoid this phenomenon. This operation is likely to improve the performance of the model and reaches the state-of-the-art effect.

E. Investigation of the Effects of Different Loss Functions

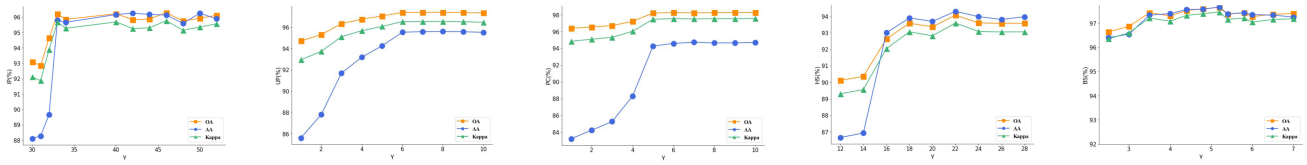
Table XV gives the experimental results of the proposed algorithm with different loss functions on the five data sets.

TABLE XV
OA COMPARISON OF LITEDEPTHWISENET BETWEEN DIFFERENT LOSS FUNCTIONS

Dataset	Standard Cross Entropy			Balanced Cross Entropy			Focal Loss		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
IP	92.13	81.56	91.03	93.66	95.46	92.77	96.29	96.14	95.77
UP	94.63	85.38	92.83	94.63	85.40	92.83	97.39	95.59	96.54
PC	97.92	92.02	97.04	97.91	91.99	97.03	98.24	94.73	97.51
HS	89.92	86.58	89.08	92.65	93.55	92.06	94.07	94.30	93.59
BS	97.40	97.26	97.18	97.84	97.90	97.66	97.59	97.52	97.39

It can be seen that the FL has a significant improvement on the performance of hyperspectral classification. Specifically, for the IP data set, due to the imbalance of sample categories, the BCEL increases by about 14% AA compared with the standard CEL, which benefits from increasing the loss weight of small class samples. Based on this, FL increases the weight of samples that are difficult to classify and further improves OA, AA, and Kappa, increasing by 2.63%, 0.68%, and 3% respectively. Compared with the growth of balanced cross-entropy loss, the performance improvement of FL obviously benefits from difficult samples. For the UP and PC data sets, there is no problem of sample imbalance, so standard cross-entropy and balanced cross-entropy show similar performance, while FL still has a significant improvement. In the UP data set, compared with the CEL and the BCEL, the OA, AA, Kappa of FL increase by about 2.77%, 10.22%, and 3.72%, respectively. The overall classification of the PC data set is less difficult and the room for improvement is small, but there is still 0.37%, 2.7%, and 0.54% increase on the three quantitative indicators respectively. On the HS data set, the FL

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 21. OA, AA, and Kappa of LiteDepthwiseNet with different values of γ on the IP, UP, PC, HS, and BS data set. (a) IP. (b) UP. (c) PC. (d) HS. (e) BS.

still contributes a 1% increase in accuracy. On the BS data set, different losses show similar results.

F. Investigation of the γ of Focal Loss

Fig. 21 shows the effect of different values of γ in the proposed LiteDepthwiseNet for OA, AA, and Kappa on the five data sets. For different data sets, the optimal weight and degree of influence are different. Of these, the value of γ has a particularly significant impact on AA because AA is particularly sensitive to the accuracy of small sample categories (often difficult-classified samples), and adjusting γ can significantly enhance attention to difficult-classified samples. In our experiment, the optimal values of γ for the five data sets IP, UP, PC, HS, and BS are 46, 8, 7, 22, and 5.2, respectively.

V. Conclusion

In this article, we propose an HSI classification network, LiteDepthwiseNet, which can achieve state-of-the-art performance with minimal parameters and low computational cost. The network is constructed by the modified 3-D depthwise convolution, which greatly reduces the model parameters and computational overhead. At the same time, this modification reduces the complexity of the model and so our model is likely to avoid the phenomenon of overfitting on data sets with very small samples. In addition, compared with the CE loss and BCE loss, we find that the FL can improve the limited performance of the model on most data sets. As a lightweight network, the proposed LiteDepthwiseNet can be deployed on devices with low computational power for HSI classification. We believe that further work can be devoted to improving the generalization ability of the model. When the distribution of the training and verification set is different, it should also achieve good performance. When the untrained data meets the category that has been seen, it should also show excellent performance, so it can be deployed in the industry to improve productivity.

Acknowledgment

The authors would like to thank Prof. D. Landgrebe for providing the Indian Pines data set, Prof. P. Gamba for providing the UP and the Pavia Center data sets, Prof. M. Crawford for providing the Botswana data set, and Prof. S. Prasad for providing University of Houston 2013 data set.

References

[1] Z. Li, L. Huang, and J. He, "A multiscale deep middle-level feature fusion network for hyperspectral classification," *Remote Sens.*, vol. 11, no. 6, p. 695, Mar. 2019.

[2] N. Yokoya, J. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated EnMAP and Sentinel-2 images," *Remote Sens.*, vol. 8, no. 3, p. 172, Feb. 2016.

[3] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[4] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[5] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[6] Q. Du and C.-I. Chang, "A linear constrained distance-based discriminant analysis for hyperspectral image classification," *Pattern Recognit.*, vol. 34, no. 2, pp. 361–373, Feb. 2001.

[7] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[8] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[9] Y. Zhong and L. Zhang, "An adaptive artificial immune network for supervised classification of multi-/hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 894–909, Mar. 2012.

[10] J. Peng, Y. Zhou, and C. L. P. Chen, "Region-kernel-based support vector machines for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4810–4824, Sep. 2015.

[11] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[12] J. Li, P. R. Marpu, A. Plaza, J. M. Bioucas-Dias, and J. A. Benediktsson, "Generalized composite kernel framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4816–4829, Sep. 2013.

[13] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[14] N. He *et al.*, "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.

[15] L. Fang, N. He, S. Li, A. J. Plaza, and J. Plaza, "A new spatial–spectral feature extraction method for hyperspectral images using local covariance matrix representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3534–3546, Jun. 2018.

[16] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[17] L. Fang, S. Li, X. Kang, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification via multiscale adaptive sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7738–7749, Dec. 2014.

[18] L. Fang, C. Wang, S. Li, and J. A. Benediktsson, "Hyperspectral image classification via multiple-feature-based adaptive sparse representation," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 7, pp. 1646–1657, Jul. 2017.

[19] J. Peng, X. Jiang, N. Chen, and H. Fu, "Local adaptive joint sparse representation for hyperspectral image classification," *Neurocomputing*, vol. 334, pp. 239–248, Mar. 2019.

[20] J. Peng, H. Chen, Y. Zhou, and L. Li, "Ideal regularized composite kernel for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1563–1574, Apr. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.

[21] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.
[22] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.
[23] X. Zhang, Y. Liang, C. Li, N. Huyan, L. Jiao, and H. Zhou, "Recursive autoencoders-based unsupervised feature learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1928–1932, Nov. 2017.
[24] B. Pan, Z. Shi, and X. Xu, "R-VCANet: A new deep-learning-based hyperspectral image classification method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1975–1986, May 2017.
[25] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.
[26] L. Fang, G. Liu, S. Li, P. Ghamisi, and J. A. Benediktsson, "Hyperspectral image classification with squeeze multibias network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1291–1301, Mar. 2019.
[27] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3322–3325.
[28] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[29] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.
[30] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[31] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.
[32] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, early access, Aug. 18, 2020, doi: 10.1109/TGRS.2020.3015157.
[33] H. Zhang, Y. Liu, B. Fang, Y. Li, L. Liu, and I. D. Reid, "Hyperspectral classification based on 3D asymmetric inception network with data fusion transfer learning," *ArXiv*, vol. abs/2002.04227, Feb. 2020.
[34] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.
[35] R. Li and C. Duan, "LiteDenseNet: A lightweight network for hyperspectral image classification," Apr. 2020, *arXiv:2004.08112*. [Online]. Available: http://arxiv.org/abs/2004.08112
[36] A. G. Howard *et al.*, "MobileNets: Efficient convolutional neural networks for mobile vision applications," Apr. 2017, *arXiv:1704.04861*. [Online]. Available: http://arxiv.org/abs/1704.04861
[37] R. J. Wang, X. Li, and C. X. Ling, "Pelee: A real-time object detection system on mobile devices," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 1963–1972.
[38] R. Ye, F. Liu, and L. Zhang, "3D depthwise convolution: Reducing model parameters in 3D vision tasks," in *Proc. Can. Conf. Artif. Intell.*, 2019, pp. 186–199.
[39] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.

Benlei Cui received the bachelor's degree from the School of Business, Changzhou University, Changzhou, China, in 2019. He is pursuing the master's degree from the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou, China.

His research interests include remote sensing image, neural network compression, and computer vision.

Xue-Mei Dong received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2002 and 2005, respectively, and the Ph.D. degree from the School of Science, Beijing University of Aeronautics and Astronautics, Beijing, China, in 2008.

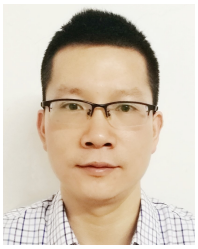
From 2008 to 2010, she was a Post-Doctoral Researcher with the Institute of Automation, Chinese Academy of Sciences, Beijing. She is an Associate Researcher with the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou, China. Her research interests include machine learning, remote sensing image processing, and data mining.

Qiaoqiao Zhan received the bachelor's degree from the College of Science, Ningbo University of Technology, Ningbo, China, in 2019. She is pursuing the master's degree from the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou, China.

Her research interests include data mining, machine learning, and hyperspectral image classification.

Jiangtao Peng received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.



Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he was a Visiting Scholar with the Department of Applied Mathematics, University of Maryland, College Park, MD, USA, with Prof. J. Benedetto, where he was involved in the study of dimensionality reduction of hyperspectral image. From 2014 to 2016, he was a Post-Doctoral Researcher with the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, where he was involved in the study of intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is an Associate Professor with Ningbo University, Ningbo, China. He has authored or coauthored more than 70 journal articles. His research interests include hyperspectral image processing with manifold learning, anomaly detection, and target recognition of remote sensing imagery.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:02:38 UTC from IEEE Xplore. Restrictions apply.